

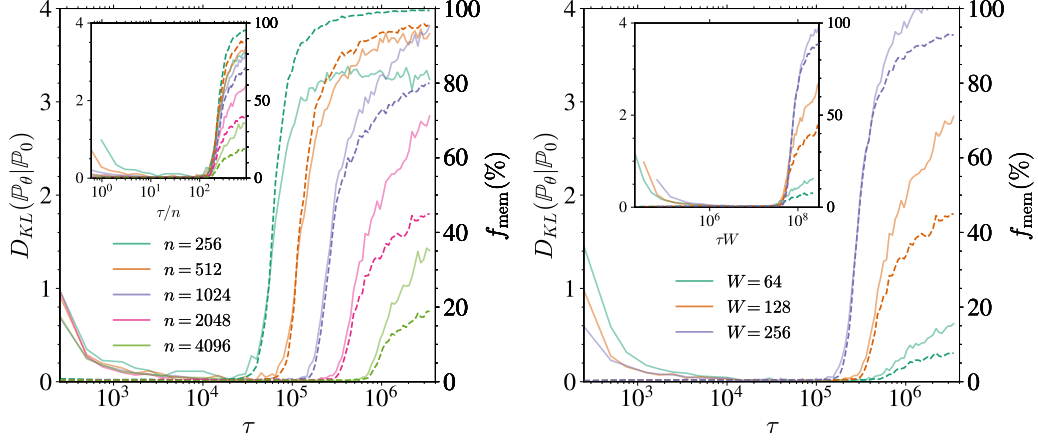


Figure 9: **Generalization–Memorization transition as a function of the training set size n and width W for ResNet score models on GMM ($d = 8$).** KL divergences (solid lines, left axis) and memorization fraction f_{mem} (in %, dashed lines, right axis) against training time τ for various (Left) $n \in \{256, 512, 1024, 2048, 4096\}$ at fixed $W = 128$. (Right) $W \in \{64, 128, 256\}$ at fixed $n = 2048$. Insets: $D_{\text{KL}}(\mathbb{P}_\theta|\mathbb{P}_0)$ and f_{mem} against the rescaled time τ/n (left) and τW (right).

with

$$I(t) = \frac{\beta_t}{2N} \sum_{\mu=1}^N [\tilde{\mathbf{x}}_\mu \mathbf{s}_\theta(\tilde{\mathbf{x}}_\mu, t) + \mathbf{s}_\theta(\tilde{\mathbf{x}}_\mu, t)^2]. \quad (28)$$

This expression assumes that the model learns an accurate representation of the score function. It is noteworthy to mention that samples are generated using standard Euler-Maruyama discretization of the backward process 2 of the MT over $T = 1000$ timesteps.

B.2 Scaling of τ_{mem} and τ_{gen} with n and W

In Fig. 9, the left panel shows how the KL divergence and memorization fraction evolve with training time τ for different training set sizes n at fixed width $W = 128$, while the right panel fixes $n = 2048$ and varies W . In both cases, we observe two distinct phases. First, the KL divergence decreases to near zero on a timescale τ_{gen} independent of n during which the model fully generalizes ($f_{\text{mem}} = 0$). Beyond τ_{gen} , both $D_{\text{KL}}(\mathbb{P}_\theta|\mathbb{P}_0)$ and f_{mem} begin to rise at a time τ_{mem} that scales linearly with n , as highlighted by the inset of the left panel. In contrast, τ_{mem} scales with W^{-1} , as shown in the inset of the right panel. While the precise dependence of τ_{gen} with W remains inconclusive in this setting and require a more careful analysis, these results on the GMM mirror the main findings of the MT: the training dynamics of DMs unfolds first in a generalization phase and only later – at $\tau_{\text{mem}} \propto n/W$ – memorization begins.

B.3 Discussion on conditional diffusion models

Conditional generation aims to sample from distributions of the form $p(\mathbf{x}|\mathbf{y})$, where $\mathbf{y}$ denotes a conditioning variable such as a class label, a text embedding, or any other contextual information. DMs can naturally be extended to this setting using for instance classifier-free guidance [21]. Although conditioning often improves sample quality in practice, memorization effects have also been observed in models trained conditionally [51, 61, 10]. Our analysis does not rely on the model being unconditional since these variables typically enter the model as additional inputs and we expect our result to hold in this setting as well. To investigate it, we train a classifier-free guidance model to generate sample from our Gaussian mixture conditionally on the class label, and compute the memorized fraction as a function of τ that we report in Fig. 10. In the inset, when rescaling the training time by n , the curves for $n \in \{256, 512, 1024\}$ all collapse perfectly, confirming that the phenomenon persists in the conditional setting. For more complex datasets, τ_{mem} and τ_{gen} may in fact depend on the conditioning variable and intermediate regimes could exist where certain classes have already entered the generalization (or memorization) phase while others have not yet.